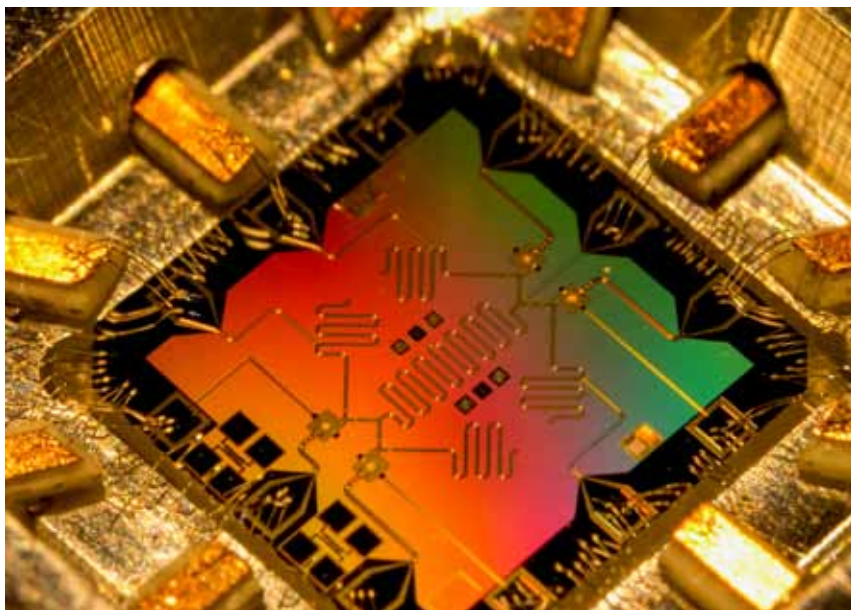




Model of the quantum processor computer

players in the field are summarised below.

1. The Canadian firm D-Wave has announced that it is commercialising its next-generation system, **Advantage**, built for businesses. The company's computer is already being used, via its cloud, by some corporations to build and run applications for their operations. Now D-Wave is making Advantage available to other businesses as well.

The system is made of more than 5000 qubits, which is more than double the count of processing units in its earlier 2000Q machine. Each of these qubits will connect with 15 others, increasing the connectivity by two-and-a-half times its predecessor's, leading to larger, denser, and more powerful graphs for building commercial applications.

2. **IONQ**, a firm in Maryland, USA, uses trapped ytterbium ions manipulated by lasers to perform the calculations.

3. **Rigetti Computing**, a Californian firm, employs microwaves to control pairs of electrons flowing through superconducting circuits.

4. **Microsoft** is working on a 'topological' quantum computer that relies on the interactions of super-cold electrons.

5. **Psi Quantum** does its computing with photons that run along waveguides etched onto ordinary silicon

chips. It hopes to leapfrog the noisy intermediate-scale quantum (NISQ) technology era entirely and produce a full-fledged quantum computer within the next five years. One of the firm's founders says that Psi Quantum is already producing wafers full of quantum-computing chips in partnership with Global Foundries, a big contract chip-maker.

6. **Q-CTRL**, an Australian start-up, has raised tens of millions of dollars from investors including Sequoia Capital and In-Q-Tel. Their engineers have used machine learning to improve hardware-control routines written initially by human beings. This can cut error rates by 90 percent and reduce variability across an individual chip by a similar amount.

7. **IBM** recently committed to having a functioning 1,000-qubit quantum computer by 2023. With the IBM Quantum Condor planned for 2023, running 1121 qubits to be exact, quantum computers would start to tackle a substantial number of genuine real-world calculations, rather than being restricted to laboratory experiments.

The director of IBM Research hopes to bring error rates down from one percent today to around 0.0001 percent in the Condor processor. It is a bold target considering IBM's biggest quantum computer to date holds just 65 qubits. The company says it plans

to have a 127-qubit machine ready in 2021, a 433-qubit one available in 2022, and a computer holding a million qubits and hopefully a functioning 1000-qubit quantum computer by 2023. One thousand qubits in total still would not be enough to take on full-scale quantum computing challenges, but it would be enough to maintain a small number of stable, logical qubit systems that could then interact with each other.

8. **Xanadu** quantum computers based on photons may possess key advantages over those based on electrons. The Canadian quantum computing start-up, Xanadu (a part of IBM's quantum computing Q-network), has, for the first time, made a photonic quantum computer publicly available over the cloud.

The classic approach to photonic quantum computing, a linear optical quantum computing, relies on qubits, each based on a single photon. This strategy manipulates photons with mirrors, beam splitters, and phase shifters. Single photon detectors are then used to help read the results of what these devices have done. The problem with this approach is that single photons are difficult to experiment with, generally limiting this strategy to a handful of photons. In contrast, Xanadu's strategy, known as Continuous Variable Quantum Computing, does not employ single-photon generators. Instead, the company relies on so-called 'squeezed states' consisting of superpositions of multiple photons.

Some new technologies that are being explored in the field of quantum computing are described below.

**Photonics.** If a stray photon, which is a particle of light, from outside the system were to interact with a qubit, its wave would interfere with the qubit's superposition, essentially turning the calculations into a jumbled mess—a process called decoherence. It turns out that superconducting cavities made of niobium, normally used to propel particle beams in accelerators, could be the solution. These cavities need to be constructed very precisely and operate at very low temperatures to efficiently propagate the radio waves that accelerate particle beams.